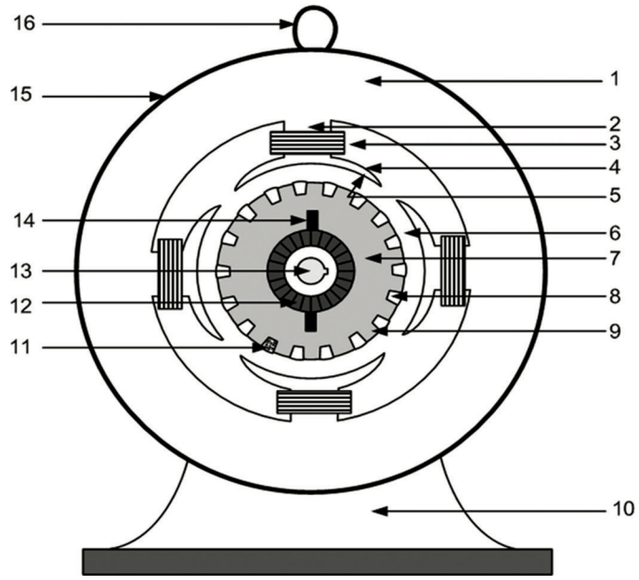


1. Yoke
2. Pole core
3. Pole winding
4. Pole shoe
5. Pole face
6. Air gap
7. Armature core
8. Slot
9. Tooth
10. Foot
11. Armature winding
12. Commutator
13. Shaft
14. Brush
15. Frame
16. Lifting leg

Fig 7.3 *Parts of DC Machines*

7.3 CONSTRUCTION OF DC MACHINES

Invariably speaking, the construction of the DC generator and DC motor are same. So any DC generator can be treated as DC motor. The construction of a 4-pole DC machine is given in the fig 7.3.

All DC machines have mainly two parts:

1) Stator (outer stationary part)
2) Rotor (inner rotating part)

The Stator consists of Yoke or Magnetic frame, poles, pole shoe, field winding.

Rotor has the following parts:
Armature – Armature core, Armature winding
Commutator – Brushes, Bearing.

7.3.1 Yoke

The Functions of Yoke are:

1) It provides mechanical support to the poles and acts as a protecting cover for the whole machine.
2) It carries the magnetic flux produced by the poles.

7.3.2 Field System

Field system consist of two main parts, poles and field winding.

i) Poles

Each pole is divided into two parts, namely:

1) Pole core
2) Pole shoe

The functions of pole core and pole shoe are:

1) Pole core carries field winding which is necessary to produce the flux.

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